

random process assignment 2

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1 Pr.1

Poisson process has independent inter arrival times.
Each inter arrival time is exponentially distributed with parameter λ :

$$\Pr[T_n > t] = e^{-\lambda t} = F_{T_n}^c(t)$$

$$f_{T_n}(t) = \frac{d}{dt}(F_{T_n}) = \frac{d}{dt}(1 - e^{-\lambda t}) = \lambda e^{-\lambda t}$$

also we know $f_{t_n} \sim \text{Erlang}(n, \lambda)$:

2 Pr.2

$$\begin{aligned} E[X] &= AE_{F,\theta}[\cos(2\pi Ft + \theta)] \\ &= \iint_{f,\theta} \frac{1}{4\pi} \cos(2\pi f't + \theta') d\theta' df' \\ &= \int_{f \in \text{supp} F} -\sin(\theta + 2\pi f't) \Big|_0^{2\pi} df' = 0 \end{aligned}$$

$$\begin{aligned} R_X(\tau) &= E[X(t)X(t+\tau)] \\ &= A^2 E[\cos(2\pi Ft + \theta) \cos(2\pi F(t+\tau) + \theta)] \\ &= A^2 E[0.5(\cos(2\pi F\tau) + \cos(4\pi Ft + 2\theta + 2\pi F\tau))] \\ &= A^2 E[0.5(\cos(2\pi F\tau))] = \int_2^4 \frac{A^2}{2} \cos(2\pi f\tau) df \\ &= \frac{A^2}{8\pi\tau} (\sin(8\pi\tau) - \sin(4\pi\tau)). \end{aligned}$$

$\{E[X] = 0 \ \& \ R_X : \tau \mapsto \mathcal{X}\} \rightarrow X$ is WSS.

3 Pr.3

$E[Y] = E[A]E[e^{jX}]$ since 'A' and 'X' are independent

$$\eta_X = 0 \rightarrow \text{Var}(X) = E[X^2] = R_X(0)$$

• *remember* $E[e^{j\omega X}] = \underbrace{e^{j\mu\omega - \frac{\sigma^2\omega^2}{2}}}_{\omega=1, \mu=0} = e^{\frac{-R_X(0)}{2}}$

• *also* $E[A] = \lambda \rightarrow E[Y] = \lambda e^{\frac{-R_X(0)}{2}}$

Now lets compute auto-correlation:

$$R_Y(t, \tau) = E[A^2]E[e^{j(X_t - X_{t+\tau})}]$$

• $\text{Var}(X_t, X_{t+\tau}) = E[X_t^2] - 2E[X_t X_{t+\tau}] + E[X_{t+\tau}^2]$
 $= R_X(0) + R_X(0) - 2R_X(\tau)$

• $E[A^2] = \lambda + \lambda^2$

$\rightarrow R_Y(t, \tau) = (\lambda + \lambda^2) \exp(2R_X(0) - 2R_X(\tau))$

as you can see Y is a WSS random process.

4 Pr.4

4.1

' because X_3, X_2, X_1 all zero-mean Gaussian their joint distributions are also zero-mean Gaussian so $E[X_3 | X_2 = x_2]$ could be another zero-mean Gaussian that could be a fraction of X_2 :

$$E[X_3|X_2] = \frac{\int_{\{X_3 X_2\}} x f_{X_3 X_2} dx}{\int_{\{X_2\}} f_{X_2} dx}$$

$$E[X_3|X_2] = \frac{E[X_3 X_2]}{\int_{\{X_2\}} f_{X_2}} \rightarrow E[X_3|X_2] \underbrace{\int f_{X_2} dx}_{E[X_2]} = E[X_3 X_2]$$

$$E[E[X_3|X_2] X_2] = E[a X_2 X_2] = E[X_3 X_2]$$

- $E[a X_2^2] = a \text{Var}(X_2)$
- $E[X_3 X_2] = \text{Cov}(X_3, X_2) = R_X(\tau)$

$$E[X_3|X_2] = \frac{R_X(\tau)}{\text{Var}(X_2)} X_2 = \frac{R_X(1-0.5)}{R_X(0)} X_2 = \left(\frac{2}{\pi}\right)^2 X_2$$

so we have $Y = \left(\frac{2}{\pi}\right)^2 X_2$. to calculate $\Pr(|X_1 + 3Y| > 1)$ we need to know $\sigma_{X_1+3Y}^2 = \text{Var}(X_1) + 9\left(\frac{2}{\pi}\right)^2 \text{Var}(X_2) + 2\text{Cov}(X_1, 3\left(\frac{2}{\pi}\right)^2 X_2) = 1 + 15\left(\frac{2}{\pi}\right)^2$ then from table we can find $\Pr(|X_1 + 3Y| > 1) = 2(1 - \Phi(1/\sigma))$ which in this case is 59%.

4.2

$$R_{Z_1 Z_1}(t_1, t_2) =$$

$$\begin{aligned} & E[(X_1 \cos(t_1) + X_3 \sin(t_1))(X_1 \cos(t_2) + X_3 \sin(t_2))] \\ &= \cos(t_1) \cos(t_2) E[X_1^2] + \sin(t_1) \sin(t_2) E[X_3^2] + \\ & (\cos(t_1) \sin(t_2) + \cos(t_2) \sin(t_1)) E[X_1 X_3] \end{aligned}$$

$$(R_X(1) = 0, R_X(0) = 1) \rightarrow R_{Z_1 Z_1}(\tau) = \dots$$

$$\cos(t_1) \cos(t_2) + \sin(t_1) \sin(t_2) = \cos(\tau) \quad (4.2.1)$$

Likewise it can be shown:

$$R_{Z_2 Z_2}(\tau) = \cos(\tau) \quad (4.2.2)$$

Now lets find cross-correlation:

$$\begin{aligned} R_{Z_1 Z_2} &= E[Z_1(t_1) Z_2(t_2)] \\ &= E[(X_1 \cos(t_1) + X_3 \sin(t_1))(X_1 \sin(t_2) + X_3 \sin(t_2))] \\ &= \cos(t_1) \sin(t_2) E[X_1^2] + \sin(t_1) \cos(t_2) E[X_3^2] + \\ & (\cos(t_1) \cos(t_2) + \sin(t_2) \sin(t_1)) E[X_1 X_3] \\ &= \sin(t_1 + t_2) = \sin(2t + \tau) \quad (4.2.3) \end{aligned}$$

as it can seen from (4.2.1), (4.2.2), (4.2.3) Z_1 and Z_2 are individually **Wide Sense Stationary** but their Cross-Correlation is not shift-invariant.

X_3, X_2 are zero-mean jointly Gaussian so $E[Z_3] = \eta_{Z3} = 0$. and for $\text{Var}(Z_3)$ we have:

$$\sigma_{Z3}^2 = t^2 \text{Var}(X_3) + \text{Var}(X_2) + 2t \text{Cov}(X_3, X_2) = \dots$$

$$\dots = t^2 R_X(0) + R_X(0) + 2t R_X(0.5) = t^2 + 1 + 2t \left(\frac{2}{\pi}\right)^2$$

So the distribution of Z_3 becomes:

$$Z_3 \sim \mathcal{N}(0, \sigma_{Z3}^2)$$

$$f_Z(z) = \frac{1}{\sigma_{Z3} \sqrt{2\pi}} e^{-1/2 (z/\sigma_{Z3})^2}$$

Variance of sample average $\tilde{X}_k = 1/n \sum X(k/2)$:

$$\text{Var}(\tilde{X}_k) = \text{Var}(1/n (X(1/2) + \dots + X(n/2)))$$

$$= \frac{1}{n^2} \sum_i \sum_j \text{Cov}(i/2, j/2) \quad \underline{\text{Cov} = R}$$

$$\rightarrow \text{Var}(\tilde{X}_k) = \frac{1}{n^2} \sum_{i,j} R_X\left(\frac{|i-j|}{2}\right)$$

if $|i-j| = \tau \quad \forall i \neq j$ then:

$$\text{Var}(\tilde{X}_k) = \frac{1}{n^2} (n R_X(0) + 2 \sum_{m=1}^{n-1} (n-m) R_X(m/2))$$

$$\dots = \frac{1}{n} + \frac{2}{n^2} \sum_{m=1}^{n-1} R_X\left(\frac{m}{2}\right)$$

5 Part B.a

Random process $N(t)$ has stationary increments.



